INCH-POUND

A-A-59572
29 March 2004

COMMERCIAL ITEM DESCRIPTION

GAGE, COMPARATOR, ELECTRIC

The General Services Administration has authorized the use of the Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

1.1 SCOPE. This Commercial Item Description (CID), details requirements for furnishing and installing a commercially available precision Roundness and Geometry Measuring System (RGMS). This RGMS will be used to determine geometric form and position of jet engine components.

2. SALIENT CHARACTERISTICS

2.1 General Requirements. Equipment covered by this CID shall be commercially available equipment capable of measuring the following parameters: concentricity, roundness, eccentricity, squareness, runout, parallelism and circumferential flatness of part features. The overall system error is not to exceed 0.000040 inches. The equipment specified herein shall be complete to function properly without any additional equipment other than electrical power, compressed air, and the environmental conditions specified in paragraph 2.5.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/TILCC, 420 Second Street, Suite 100, Robins AFB, GA 31098-1640

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2.2 General Operation. Operationally, a part shall be positioned on the turntable. Usually, two electronic gage heads, supported by appropriate stands, shall be positioned to indicate selected features as the table is rotated. The gage heads shall provide electronic signals, which are amplified and fed directly to a processing data module, which in turn will cause the origination of a polar plot chart. This chart will then be analyzed to determine the condition of the feature in question. This system will be used to determine part feature characteristics such as concentricity, roundness, eccentricity, squareness, circumferential flatness runout, and parallelism of component parts.

2.3 Safety. The RGMS shall be equipped with all necessary safety devices for protection against electrical shock, and to insure the safety of the operator. If the RGMS is equipped with electronic/electrical interlocks, they shall be clearly marked in a fully visible location.

2.4 Accuracy.

2.4.1 System Accuracy. The overall system error shall not exceed 0.000040 inches. The RGMS shall be capable of producing identical traces when a second trace is run over the same test piece without changing any amplifier or stylus positioning adjustment. Any difference in the two traces shall be added to all other errors when determining the system accuracy. Other accuracies specified in this paragraph are listed as intermediate accuracies for intermediate testing and shall not be considered when arriving at the “system accuracy”.

2.4.2 Spindle Accuracy. Axial error shall not exceed 0.000010 inches. Total indicator runout (TIR) radially shall not exceed 0.000010 inches. Coning above the turntable shall not exceed 0.0000010 inches per inch of vertical distance.

2.4.3 Worktable Accuracy. Radial flatness shall be within 0.00020 inches. Circular flatness shall be within 0.00010 inches.

2.4.4 Granite Table Accuracy. The granite table shall be flat within tolerances established in GGG-P-463c for Grade B surface plates.

2.5 Operating Environment. The following capabilities/characteristics exist in the location where the RGMS will be installed. The supplier of the RGMS shall verify that the following list of available services are adequate and shall also assure the Government in writing that operating environment will not degrade the accuracy and operation of the machine.

2.5.1 Electrical Power. 110 VAC +/- 10% is available.

2.5.2 Compressed Air. 90psi +/- 10% with a flow rate of 10 SCFM +/- 10%. Quality of the compressed air is unknown.

2.5.3 Vibration. Vibration in the area proposed for installation of this machine is estimated at:

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FREQUENCY	AMPLITUDE (Peak to peak all directions)
100 Hz	0.000006 inches
1000 Hz	0.0000002 inches

2.5.4 Temperature. Ambient air temperature of the location where the RGMS will be installed varies between 60 to 80 degrees F.

2.5.5 Humidity. Relative humidity of the location where the RGMS will be installed varies between 40 to 80 percent.

2.6 General Design.

2.6.1 Design Currency. The RGMS shall be of the manufacturer's most current proven commercial design providing the performance capabilities specified herein as minimum requirements. The machine shall be designed to perform all of the measuring functions specified herein.

2.6.2 Operating Controls. All operating controls and all manual adjustments shall be grouped and located for the operator's convenience and shall be clearly marked as to functions. All electronic alignment controls shall be provided with appropriate locking devices and means for making fine adjustments.

2.7 Construction. All parts and materials used in the construction of the RGMS components shall be new. The quality shall be such as will provide the strength, rigidity, and wear resistant characteristics necessary to meet the standards for construction, accuracy, performance and safety requirements specified herein. All parts and mechanisms subject to periodic adjustment, replacement, or servicing shall be readily accessible. The system shall be complete so that when installed and connected to the electrical power supply and compressed air supply, it will be ready for use. The machine shall be supplied with the necessary equipment for filtering, drying and regulating the pressure of the air to be used. Construction shall be free from any characteristics or defects that will prevent the machine from passing any of the examinations and checkouts required in Section 4, Quality Assurance Provisions.

2.8 Mechanical System Construction.

2.8.1 Bearings. All rotating machine components shall be equipped with anti-friction and free-floating air, ball or roller bearings as specified by the style required. All bearings shall be specially constructed precision ground and lapped for withstanding the applicable radial and axial loads. These bearings shall be adequately identified for replacement purposes. All sliding surfaces of the machine that are subject to wear shall have provisions for lubrication and sha11 have replaceable bearing surfaces.

2.8.2 Bearing Supports. All bearing supports shall be designed and constructed to eliminate inaccuracies caused by deflections of machine components and to prevent applicable factors from adding total error.

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2.8.3 Moveable Surfaces. All moving parts shall be equipped with replaceable bearings. All bearings that are not of the sealed-for-life or free-floating air type shall be equipped with an automatic lubrication system. Re-circulating lubricating oil systems shall be equipped with a filter. Each oil reservoir shall have a sight device for indicating safe oil level. All oil holes and grease fittings shall be easily accessible and clearly marked.

2.8.4 Castings and Forgings. All castings and forgings shall be free of defects, scale and mismatching. No process such as welding, peening, plugging, or filling with solder or metallic paste shall be used for reclaiming any defective part for use in the machine.

2.8.5 Welding, Brazing or Soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

2.8.6 Surfaces. All surfaces of castings, forgings, stampings, molded and/or welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials which could affect the operation, accuracy, and life expectancy of the equipment. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's commercial practice.

2.8.7 Fastening Devices. All screws, pins, bolts and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts when subject to removal or adjustment shall not be swagged, peened, staked, or otherwise permanently deformed.

2.8.8 Painting. All non-working, non-moving parts shall be protected from corrosion in accordance with the manufacturer's standard commercial practice. All non-working. non-moving, non-chrome exterior part of the machine shall be painted in a suitable finish and color of manufacturer's option.

2.8.9 Instruction and Identification Plates. All instruction plates and identification plates attached to the system components shall be of a corrosion resistant material. All wording shall be permanent and in the English language using plain boldface letters with a contrasting background.

2.8.10 Identification Plate Marking. The markings on identification plates shall include Nomenclature, Manufacturer's Name, Manufacturer's Model Designation, Manufacturer's Serial Number, and Power Input. Contract Number, National Stock Number, and Date of Manufacture. The markings shall be in accordance with MIL-STD-130K.

2.8.11 Maintenance Identification Plates. Instruction plates shall be furnished to indicate points, recommended time interval, type, viscosity and number of lubricants to be used. Where it is applicable, the capacity and pressure shall be specified on an instruction plate for compressed air inlets.

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2.9 Electrical/Electronic System Requirements

2.9.1 Electrical System Wiring. The electrical systems shall be wired, enclosed, fused and grounded in accordance with National Electrical Code (NEC). All motors shall have thermal overload protection. The machine shall be operable from 110 VAC +/- 10%, 60 Hz, single phase, alternating current electrical power.

2.9.2 Electrical Components. All electrical components on the measuring machine including controls, relays, switches, overload protection devices, ground fault interrupters and their wiring and enclosures shall conform to and shall be located in accordance with NEC. No component part shall be used in any application where the requirements exceed the manufacturer's rating for the part.

2.9.3 Electrical Connections. Electrical connections of each item comprising the RGMS shall be mechanically secured, soldered, or brazed in accordance with NEC.

2.9.4 DC Power Supply. If any requirement for direct current (DC) power exists, the manufacturer shall include a standard commercial DC power supply integrated into the machine/system. This power supply shall be rated, as a minimum, for one and one-half (1-1/2) times maximum demand.

2.9.5 On/Off Switch. A separate easily accessible switch shall be provided which controls all AC power supplied to entire RGMS.

2.10 RGMS System. The RGMS system shall consist of the following.

2.10.1 Rotary Table (Spindle). Spindle shall be stainless steel, heavy duty, air-bearing design with the diameter of at least 10.6 inches. The spindle shall have axial load capacity of at least 1000 pounds. The following minimum requirements for spindle shall be met:

1) Axial Stiffness	4.5 pounds per micro inch
2) Radial Accuracy	0.00001 inches
3) Axial Accuracy	0.00001 inches
4) Coning Accuracy	0.000001 inches per inch

2.10.2 Tilt/Center Worktable (Turntable). The worktable shall rotate on an air-bearing spindle. The air-bearing spindle shall conform to the safeguards in paragraph 2.13.10.1 and 2.13.10.2. The worktable shall be either motor driven or rotated by hand when the motor is not engaged. (When compressed air is not supplied to the system, the worktable shall NOT rotate.) The worktable shall rotate at a constant speed of two (2) RPM when the motor is engaged. The

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worktable shall have circular holding area that is no less than 24 inches in diameter. Worktable flatness shall conform to ASME Y14.5M -1994. The worktable shall support a balanced load of (at least) 1000 pounds. This load shall not degrade the accuracy beyond requirements in either static or dynamic operating conditions. (Refer to paragraph 2.4 for accuracy requirements.) The worktable shall have four knobs: two for leveling and two for centering. These knobs shall be located 90 degrees apart from each other. The worktable shall have a manual tilting range of no less +/- 1 degree (2 degrees total). Zero degree is defined as when the surface of worktable is parallel to the granite table. The axis tilt control shall be readily and easily accessible to the operator. The tilt adjustments used for leveling the worktable shall be equipped with mechanisms that shall function properly under conditions of maximum rated balanced load. "Function properly" is defined as follows: Mechanisms shall respond linearly and shall NOT stick, bind, or wring under full load conditions. Worktable shall have a manual centering range of no less than +/- 0.100 inches (.200 inches total). Centering controls shall be readily and easily accessible to the operator. Under conditions of maximum rated balanced, the centering adjustments shall function properly. Worktable shall have centering grooves, which are nominally concentric with axis of the worktable. Centering grooves may be spaced any; distance apart up to one (1) inch, however, the distance between all centering grooves shall be equally spaced. Worktable shall have tapped holes, which will be used to hold down fixtures/parts. Tapped holes shall be arranged in at least three (3) equally spaced (120 degrees nominal) radial rows. Holes within each specific row shall be arranged equally spaced and no farther than two (2) Inches apart. Hole spacing shall be the same for all rows. Hole size shall be l/4"-20. Worktable shall be constructed of stainless steel. Worktable shall be installed in a granite table so the height of the top of worktable shall be no greater than 12 inches above granite table. Worktable shall be positioned on the granite table according to the following criteria:

1) Location shall be as close to the edge of the longer side as possible to provide maximum operator utilization of the table.

2) Worktable shall be positioned far enough away from the edges of the granite table so that maximum loaded worktable/spindle weight will not deform the granite table beyond flatness tolerances as specified for Grade B in Federal Specification GGG-P-463C paragraph 3.3.4.

2.10.3 Granite Table. Granite table shall be made from monolithic high density, fine grain black granite (diabase). Its dimension shall be 5 feet long and 3 feet wide. The thickness of the granite table shall be at least 10 inches. The granite table shall be designed and constructed to support a total normal load of 1000 pounds plus the entire weight of the worktable and spindle. The granite table shall have a steel base with four (4) shock mounts. These shock mounts shall be adjustable in vertical direction and equipped with locking devices to preclude vibration movement once adjustment is made. Workmanship of work surface on the granite table shall conform to Federal Specification GGG-P-463C, paragraph 3.3.1. Surface texture of the granite table shall be in accordance with Grade B requirements in Federal Specification GGG-P-463C, paragraph 3.3.2. Tolerances on repeat reading measurement of the granite table shall be in accordance with Federal Specification GGG-P-463C, paragraph 3.3.3 and Table n (Grade B

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Column). Flatness tolerance of the granite table shall conform to Grade B requirements in Federal Specification GGG-P-463C, paragraph 3.3.4.

2.10.4 Gage Posts. Two (2) free-standing gage posts shall be provided. The length of gage posts shall be at least forty-eight (48) inches in height. Gage posts shall be rack-and-pinion drive type. Gage posts shall have arm adjustment and gage head mount fixture, permitting use of gage head in any position relative to the arm. The bases of gage post shall be made of rigid cast iron with hardened steel pads. Bases shall be designed and constructed to avoid tipping. Gage posts shall be equipped with provisions to use two (2) arms at different levels at the same time.

2.10.5 Gage Arms. Two (2) interchangeable, friction-drive arms shall be furnished. Their length shall be twenty-four (24) inches long. These arms shall be mounted as to permit 360 degrees rotation in the vertical plane.

2.10.6 Gage Heads. Two (2) gage heads shall be provided. Gage heads shall be pivot lever transducer type, which produces linear electrical signals directly proportional to stylus displacement. Gage heads shall be compatible with the electronics system and the gaging posts of RGMS. Each gage head shall have a cable of at least 14 feet long. The following gage head criteria shall be met:

1) Probe tip to be swiveled along a minimum 180 degree arc.

2) Gage head to enter a bore no larger than 1.5 inches in diameter.

3) Stylus displacement shall be selectable for either positive or negative direction by operation of a simple reversing lever.

4) Gauging force shall be no more than five (5) grams and bi-directional.

5) Rotational axis of gage head shall use a pivot-bearing-and-slip-clutch design to ensure stability and repeatability during setup and measurement.

6) Gage heads shall be designed to facilitate rapid changing of styli by an operator. Any and all tools required to change Styli shall be provided.

7) Gage head shall have a linear range from (-0.010 inch) to (+0.010 inch).

2.10.7 Stylus. Two (2) carbide styli shall be provided. Styli shall be straight and at a minimum of 1 inch long. The contact ball shall be no greater than 0.062 inches in diameter.

2.10.8 Data Processing Module. Data processing module for RGMS shall be equipped with the most current processor/co-processor, 1.4 MB floppy disk drive, a printer and metrology software. Viewing screen shall be least 10.4 inches. Metrology software shall be user-friendly and shall perform the measurements specified in section 2.2 General Requirements. Measurement results can either be printed or downloaded onto a floppy disk. Printer shall be the most current DeskJet HP model. All cables (power, printer, interface, etc.) which are necessary to make the system

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operational and complete) shall be at least ten (10) feet long. The data processing module shall perform the following functions:

2.10.8.1 Center Polar Trace. By operator selection, data processing module shall automatically center the polar trace of a part that is located eccentric to the spindle axis by no less than one and one-half (1-1/2) times the half (1/2) scale meter range.

2.10.8.2 Least Mean Squares. By operator selection, the data processing module shall automatically chart a "least mean squares" best fit circle representing the true location of an eccentrically placed part, eliminating profile detail, for determination of true part center with respect to the spindle axis. No more than one turntable revolution shall be allowed for data taking before charting a circle. The best fit circle shall have the correct mathematical relationship to the “actual” polar trace of the feature being measured. That is, after placing an actual trace of the feature surface on the polar chart, the best fit circle may then be selected, and after one revolution, the best fit circle (without changing pen zero, electrical zero controls, etc.) will be charted automatically so that the area between the actual profile trace and the best fit circle will be equalized inside and out.

2.10.8.3 Tracing Uncentered Profiles. By operator selection, the data processing module shall permit tracing uncentered profiles, independent of correction.

2.10.8.4 Maximum Inscribed Circle. By operator selection, data processing module shall automatically chart a “maximum inscribed circle”. The maximum inscribed circle is described as one (1) circle, which is the largest possible circle inscribed within an “actual” polar trace of a feature. No portion of the inscribed circle shall cross any boundary defined by the actual trace. No more than one turntable revolution shall be allowed for data taking before charting a maximum inscribed circle.

2.10.8.5 Minimum Circumscribed Circle. By operator selection, the data processing module shall automatically chart a “minimum circumscribed circle”. The minimum circumscribed circle is described as one (1) circle, which is the smallest possible circle circumscribed outside an “actual” polar trace of a feature. No portion of the circumscribed circle shall cross any boundary defined by the actual trace. No more than one turntable revolution shall be allowed for data taking before charting a minimum circumscribed circle.

2.10.8.6 Minimum Radial Separation. By operator selection, the data processing module shall automatically chart a “minimum radial separation”. This minimum radial separation is described as two (2) concentric circles that have the least radial separation and yet contain between them all of the actual trace. No more than a total of five (5) turntable revolutions shall be required to take data and plot both circles.

2.10.9 Amplifier System. The amplifier system shall have a digital analog meter. This meter shall read zero (0) in the center with minus on the left and plus on the right. A separate “Knob” type fine adjustment shall be provided for electrically zeroing the meter for each channel/gage head. The amplifier system shall have two (2) separate gage head signal inputs with separate calibration adjustments. The amplifier shall have the following operator selectable outputs to both meter and polar chart:

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(1) Algebraic sum of two (2) input signals.

(2) Algebraic difference of two (2) input signals.

(3) Selection of either individual input signals.

(4) Individual reversible polarity of either input signal.

2.10.9.1 Ranges and Resolutions. The following operator selectable ranges and resolutions shall be provided:

RANGE	**RESOLUTION**
+/- 10,000µ”	10,000µ”
+/- 5,000µ”	500µ”
+/- 1,000µ”	100µ”
+/- 500µ”	50µ”
+/- 200µ”	20µ”
+/- 100µ”	10µ”
+/- 50µ”	5µ”

2.10.10 Compressed Air System. The compressed air system shall incorporate an air filtering and pressure relief system.

2.10.10.1 Air Filtering. The air system shall filter incoming compressed air into the quality of air best suited for the RGMS. When the filter system does not adequately purify the quality of air to be consumed by RGMS, RGMS shall automatically shutdown and give a continuous visual indication of reason for shutdown. In no case shall failure to maintain/clean the filter system cause harm to the RGMS.

2.10.10.2 Air Pressure Control. To prevent damage and/or loss of accuracy (see paragraph 2.2), the RGMS shall be equipped with automatic “shut-down” and pressure-relief system. When supplied air pressure is lower than RGMS's operating range, RGMS shall automatically shut down. When supplied air system is higher than RGMS's operating range, pressure relief system shall be automatically activated to discharge excessive air; however, this operation shall not allow pressure to drop low enough to activate the “low-pressure” shut-off system. In the event that the pressure relief system should fail and air pressure approaches an intolerable limit without relief, the RGMS shall automatically shut down. Whenever RGMS automatically shuts down, continuous visual indication of reason for “shut-down” shall be prompted. Under no circumstances, either low pressure or high-pressure supplied air shall cause harm to the RGMS.

2.11 Maintenance.

2.11.1 Maintenance Test Panel/Points. The RGMS shall be provided with a built-in test panel or test points for testing components and circuitry. In areas where it is practical to provide visual

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indication of failure in a component or subassembly of the control, suitable indicators shall be provided.

2.11.2 Maintenance Data. Two (2) complete copies of standard maintenance manuals and related documents normally furnished by the manufacturer shall be supplied with the RGMS. These manuals shall also include current and accurate circuit diagrams and schematics, which fully describe functions and symptoms indicated by the previously specified "Maintenance Test, Panels/Points". All of these documents shall be provided to the Government at the time the RGMS is installed.

2.11.3 Component Interchangeability. To provide for replacement of worn parts, all parts shall be produced or controlled by the original equipment manufacturer and being the same part number shall be dimensionally identical within manufacturing tolerance limits in use by the manufacturer.

2.12 Operating Manual. Two (2) complete copies of standard operating manuals shall be provided. These manuals shall be furnished to the Government at the time the RGMS is installed.

2.13 Special Tools. Special tools shall be kept to a minimum. Any special tooling shall be supplied with the equipment, and provisions shall be made to secure such tooling or store with the equipment to prevent loss. Special tooling is defined as that tooling necessary to operate the system in providing ranges and capacity applicable to the system requirements as specified and any other necessary tooling that is not normally commercially available.

3. REGULATORY REQUIREMENTS.

3.1 Recycled recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Responsibility for Inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Product Conformance. Prior to delivery to final destination. supplier shall perform a quality conformance inspection on the fully assembled RGMS at his own or any other facilities suitable for this inspection, unless such facilities are disapproved by the Government. After installation is complete at final destination, quality conformance inspection, which shall be observed and verified by Government representatives, shall be performed on the fully assembled and fully operational measuring machine. Complete installation (turn-key) shall be accomplished by the supplier. Failure of the measuring machine to pass the Government observed examination of one

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or more criteria shall be the cause for rejection. Quality conformance inspection shall consist of examination (paragraph 4.3) and checkouts (paragraph 4.4).

4.3 Examination. The RGMS and equipment shall be examined for design, construction, materials, components, electrical and electronic equipment and workmanship to determine compliance with the requirements of this specification. In the interest of utilizing new or improved products that may become available, the manufacturer may with prior approval from the Government, change the type of material or components supplied with this machine. Any unapproved deviation from these requirements shall be the cause for rejection.

4.4 Checkouts.

4.4.1 Worktable Flatness. Circumferential flatness shall be verified by indicating the rotating worktable at least one time between each concentric centering circle. Circumferential flatness shall meet or exceed the requirement specified in paragraph 2.2 Accuracy. Radial flatness shall be verified by radially tramming the worktable in at least six (6) randomly selected azimuths. Radial flatness shall meet or exceed the requirement specified in paragraph 2.2 Accuracy.

4.4.2 Ball Spindle Checkout. A precision test ball shall be provided by the vendor. The sphericity of the ball shall be within 0.000002 inches. This ball shall be used to verify spindle accuracy.

4.4.2.1 Radial Accuracy Checkout. To determine radial accuracy of the spindle, a trace is made at the approximate equator of the ball. This ball is then indexed 90 degrees, re-centered, and a second trace taken on the- same chart. Out -of -round in the ball is separated from spindle error by noting if any irregularity of the trace shifts 90 degrees. Any irregularity, which has not shifted 90 degrees. is attributable to TIR radial error. Radial error shall be no greater than is specified in paragraph 2.2 Accuracy.

4.4.2.2 Coning Accuracy Checkout. To determine coning accuracy of the spindle, repeat paragraph 4.4.2.1 with the test ball at no less than three (3) different heights above the worktable. Coning error shall be no greater than is specified in paragraph 2.2 Accuracy.

4.4.2.3 Axial Accuracy Checkout. To determine axial accuracy of the spindle, center the test ball within 0.000005 inches on the center of the turntable. Place the contact point of the gage head at the top dead center of the ball. Make a trace with the worktable motor driven. Any deviation from a true circle is axial error. Axial error shall be no greater than is specified in paragraph 2.2 Accuracy.

4.4.3 Circle Verification. All circles generated by the data processing module shall comply with the "System Accuracy" specified in paragraph 2.2. These checkouts will be performed using parts, supplied by the Government. Prior to this series of checkouts, the Government will verify the dimensions and geometric shape of this specific part. Circles that are to be verified include: 1) Least Squares Circle, 2) Maximum Inscribed Circle. 3) Minimum Circumscribed Circle, and 4) Minimum Radial Separation Circle.

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4.4.4 Computer Centering. The test ball previously mentioned shall be eccentrically affixed to the worktable within the limits specified in paragraph 2.13.8.1, and a centered chart is produced. The imprint on this chart shall meet or exceed "System Accuracy" as specified in paragraph 2.2.

4.4.5 Tilt and Centering. Under maximum rated load conditions the response. as specified in paragraphs 2.13.2, for "proper function" shall be verified.

4.4.6 Air Failure. The RGMS shall be in “normal” operating condition, and compressed air will be removed. Under these conditions the RGMS shall respond in the manner specified in paragraph 2.13.10.2 Air Pressure Control. This procedure shall be repeated three (3) times.

4.4.7 Granite Table. The vendor shall provide all equipment for required checkouts and examinations. The granite table shall not have any of the visible defects specified in GGG-P-463C. paragraph 4.4.1 Visual Examination. The granite table surface texture shall meet or exceed the requirements of GGG-P-463C. paragraph 4.5.4 Surface Texture. Flatness of the granite table shall conform to GGG-P-463C. paragraph 4.5.6 Acceptance Tests for Flatness. It is understood that flatness checks in the immediate worktable/spindle area cannot be performed.

4.4.8 Amplifier Checkout. Set the amplifier range switch to 0.0001. Wiring the step block on top of the worktable. While indicating the top step of the step block, adjust the amplifier to the zero position. Slowly rotate the worktable and allow the stylus to drop to the lower step. The amplifier meter should move ten (10) units- Calibrate the system as required to achieve this ten (10) unit/line movement. Relocate the step block to three (3) other random positions and repeat each of the above steps without any calibration or adjustment to the system, recording each checkout. Checkout results shall be within the tolerance of system accuracy as stated in paragraph 2.2.

5. PACKAGING.

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of Documents.

6.1.1 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

National Electrical Code (NEC).Copies of NEC standards can be obtained from UL Corporate Headquarters, 333 Pfingsten Road, Northbrook IL 60062-2096. Phone: 1-847-272-8800. Fax 1-847-407-1395.

6.1.5 Code of Federal Regulations (CRF). Copies of the CFR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

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6.1.6 ASME standards. Copies of ASME standards can be obtained from ASME, Three Park Avenue, New York, NY 10016-5990.

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

(Project 5220-0265)

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